

Keeper — Lepton Sector State Ledger (authoritative consolidation of the two-day arc)

Keeper | 2026-07-24 Fri | Single-reference tier ledger. Supersedes the scattered K853–K880 for state-of-play. The frontier is now ONE computation (Grace’s FK Wyler 3×3) — this ledger is the honest picture around it, not a new audit finding.

The one-line state

The entire charged-lepton mass sector is one Toeplitz operator on one domain with every input forced or discretely-anchored — no continuous knob — and the verdict is one banked FK Wyler diagonalization. It currently leans structural (the only hand-computable reading misses $91\times$), but the setup is airtight and the FK crank is the real test.

Tier ledger

Item	Status	Basis
Why exactly 3 generations, no 4th	BANKED / DERIVED (structure)	rank+1 = 3, coordinate-independent; T2517 $\cap$ (Wallach set T1829 or KW support-flag F86)
The mass hierarchy EXISTS (qualitative)	BANKED / DERIVED (structure)	condensate = singular Shilov-boundary measure $\rightarrow$ rich Toeplitz spectrum; bounded symbol can’t (Elie, K880)
The mass OPERATOR	FORCED	Toeplitz T_φ on $H^2(D_{IV^5})$, multiplication, full-rank (F585/F681)
The condensate LOCUS	FORCED	$\nu_R \nu_R$ Majorana on chargeless S^4 (F583, banked T2524)
The condensate SHAPE	FORCED	K-invariant Szegő measure (symmetric vacuum on symmetric sphere), F582/F682
The generation SUBSPACE (dim 3)	FORCED	rank+1 = 3 (the banked structure)
The overall SCALE (amplitude)	~DERIVED (Lane 3)	α -tower $m_e \sim 6\pi^5 \alpha^{12}$ (F402 prefactor + F426 exponent; only per-step- α identified, K871)
The matrix-element machinery	BANKED	Wyler boundary integral over $S^4 \times S^1$ (= the α integral)
Which 3 states span the subspace	DISCRETE ANCHOR (target-innocent)	anchored at banked electron $k=1$ (Grace); not chosen
The mass VALUES (1:207:3477 + PMNS)	LEANS STRUCTURAL, PENDING	the FK Wyler 3×3 diagonalization; naive α -ladder misses $91\times$ (Grace)

What’s genuinely DERIVED and bankable now (independent of the FK verdict)

1. Exactly three generations, no fourth — the durable why-3 result. Coordinate-independent.
2. The leptons are hierarchical BECAUSE the condensate is a singular boundary measure — a value-independent structural cause (Elie). Explains why every bounded/smooth reading (α -ladder, residues) failed in principle.
3. The whole sector is one forced operator — no free continuous parameter anywhere. This itself is a result: a genuine derived-vs-fit setup with all seven gates committed blind.

What's OPEN — the single frontier

The FK Wyler 3×3 diagonalization (Grace's lane, banked book machinery, NOT to be fabricated): - Compute $M_{ij} = \langle \text{stratum}_i | \text{Szego-}\varphi | \text{stratum}_j \rangle$ on the 3-dim subspace (states anchored at electron $k=1$). - Diagonalize $\rightarrow$ does it give 1:207:3477 + PMNS with nothing tuned? $\rightarrow$ charged leptons DERIVE (major) or structural, stated plainly. - Consistency check the crank must pass (do NOT pre-decide): the banked electron is a $k=1$ boundary/Shilov state and is the lightest; earlier pictures put the heaviest (tau) at the Shilov boundary. The FK 3×3 must resolve which end of the spectrum the boundary is. A genuine internal check.

PAPER (F676) state

- CONDITIONAL PASS ~90% (K876). Ships the COUNT (why-3, no-4th) — durable, coordinate-independent.
- Two fixes to reach PASS: (a) reframe the “coincidence” as the structural identity $\rho_2=a/2$ [Keeper drop-in written]; (b) foreground the KW support-flag (F86) as the primary why-3 (coordinate-consistent with the banked electron $k=1$), Wallach set = rep-theory companion [K880]; correct “electron = continuum” $\rightarrow$ “electron = $k=1$ boundary state.”
- Per-generation placement pending the FK 3×3 . Count + no-4th + Principle #16 ship regardless.

Discipline held (the two-day through-line)

Nothing false banked across ~9 muon pictures. Over-reaches caught in BOTH directions, each within the round: Grace's “electron=continuum” + eigenvalue-inversion, Lyra's rank-1 (F677) + coordinate mislabels, Elie's F677 wall + ZZ/WW numerology (C11), and the auditor's own — K835 Pin⁺, K852 sub-domain, K859 over-negative, K872/K873 coordinate patches, K878 O7 rep-theory drift — all self-corrected, several via Casey's repeated “linear algebra, one D_IV⁵” steer. The steer, applied three times, removed every continuous fit-hiding place.

The honest frontier (one sentence)

The lepton sector is a fully-forced one-operator setup whose qualitative results (three generations, hierarchy-exists) are banked, whose values lean structural pending a single banked FK integral no one will fabricate, and whose paper ships the durable count once reframed onto the KW support-flag.

— Keeper Lepton Sector State Ledger, 2026-07-24. Authoritative consolidation. Banked: why-3/no-4th (structure), hierarchy-exists (singular boundary measure, Elie), the forced one-operator setup. Open: the FK Wyler 3×3 verdict (leans structural, naive $91 \times$ off). Paper: count ships, foreground KW support-flag (F86), placement pending. See [[Keeper_K880_paper_thesis_tension_banked_electron_k1_not_the_Wallach_continuum_foreground_the_KW_support_07-24]], [[Keeper_K879_Casey_steer_dissolves_my_own_O7_the_three_states_ARE_the_three_forced_strata_mass_matrix_is_a_07-24]], [[Keeper_K871_CORRECTION_to_K870_Lane3_scale_is_NEARLY_DERIVED_not_the_bottleneck_F402_prefactor_F4_07-24]].